

Machine Learning III

Digitization and Data Analytics

Structure of the Lecture Series

Lecture 1: Introduction

Lecture 2: Data Preprocessing

Lecture 3: Research Data Management

Lecture 4: Data Analytics and Statistics

Lecture 5: Machine Learning I

Lecture 6: Machine Learning II

Lecture 7: Machine Learning III

Lecture 8: Deep Learning I

Lecture 9: Deep Learning II

Lecture 10: Distributed Computing I

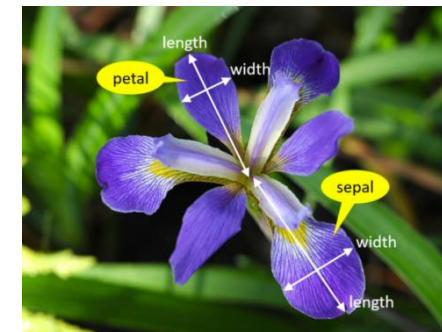
Lecture 11: Distributed Computing II

Lecture 12: Consequences

Previous Lectures on ML

- Unsupervised and Supervised Learning
- Unsupervised Learning
 - Similarities and Distances
 - Clustering: Partitioning, Hierarchical clustering, Density-based, Assessment of clustering results
- Supervised Learning
 - Training and Learning
 - Classification: Decision trees, Bayes Classification Methods
 - Model Evaluation and Selection
- Iris dataset

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
4.9	3.1	1.5	0.1	setosa
5.1	3.8	1.5	0.3	setosa
4.7	3.2	1.6	0.2	setosa
5.1	3.4	1.5	0.2	setosa
5.0	3.3	1.4	0.2	setosa
5.2	2.7	3.9	1.4	versicolor
5.7	2.6	3.5	1.0	versicolor
5.7	2.8	4.1	1.3	versicolor
6.0	2.2	5.0	1.5	virginica



<https://www.integratedots.com/determine-number-of-iris-species-with-k-means/>

Structure of this Lecture

1. Dimension Reduction (unsupervised)
 1. Principal component analysis (PCA)
 2. t-distributed stochastic neighbor embedding (t-SNE)
 3. Autoencoder
2. Linear Regression (supervised)
 1. Model, estimation
 2. Goodness of fit
 3. Variable transformation
 4. Model reduction technique
3. Logistic Regression (supervised)
 1. Model, estimation
 2. Goodness of fit

1. Dimension Reduction

Aim:

- transform the available data X to data of a lower dimension without losing (much) information.
- That means, to find a function $f: X \rightarrow Y$, optimal in some sense.

Why:

- visualization: e.g., 10D vs. 2D;
- **noise reduction**: mistakes/outliers occur in data, often in only one dimension;
- computational efficiency: e.g., the inverse of (5×5) matrix vs. (2×2) ;
- **storage** memory;
- improved **accuracy**.

Example: Subjects' grades

- Let X be a table with grades of 50 students for seven subjects
 - math, algebra, programming, statistics,
 - English, philosophy, history.
- (50×7) -dim. data.

N	math	alg	Stat	prog	En	phil	hi
1	88	75	80	95	56	48	63
2	46	42	37	51	98	90	72
3	61	49	51	82	75	76	85
...							
50	73	81	95	78	76	80	69

- Idea: students good in one subject should be good in others, too.

Example (cont.)

Assume that one hidden factor, e.g. IQ, influences linearly all grades of student i .

- $\Rightarrow$ All subjects are linearly correlated.
- $\Rightarrow$ The grades in different subjects do not contain completely new information.
- $\Rightarrow$ For the student i , each subject j

$$\text{grade}_{ij} \approx a_j + b_j \cdot \text{IQ}_i$$

With

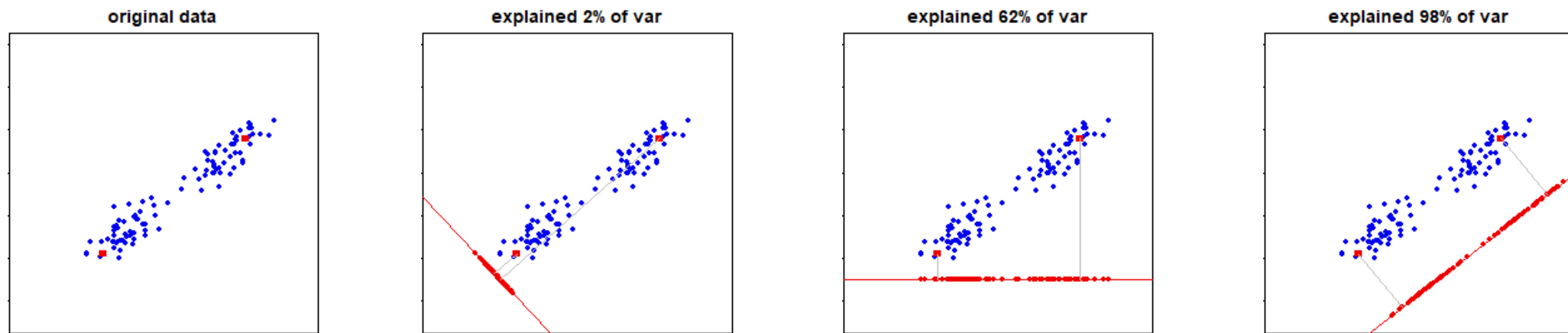
- a_j - subject-specific shift/intercept,
 - b_j - subject-specific slope/weight,
 - IQ_i - latent ability of student i .
- $\Rightarrow$ The data X can be represented by one value per student

$$X \in \mathbb{R}^{50 \times 7} \rightarrow \text{IQ} = (\text{IQ}_1, \dots, \text{IQ}_{50})^T \in \mathbb{R}^{50 \times 1}$$

1.1. PCA: Idea

Aim:

- Transform the data X to a data of a lower dimension without losing (much) information using a linear transformation.
- Mathematically, find β with $||\beta|| = 1$, so that for $Y = \beta X$ satisfies $\sum \text{var}(Y_j) \rightarrow \max$



- Questions: Is the slope important? Is the shift important?

Eigenvalues and Eigenvectors

- Empirical covariance matrix $\text{cov}(X) = \frac{1}{n} (X - \bar{X})'(X - \bar{X})$

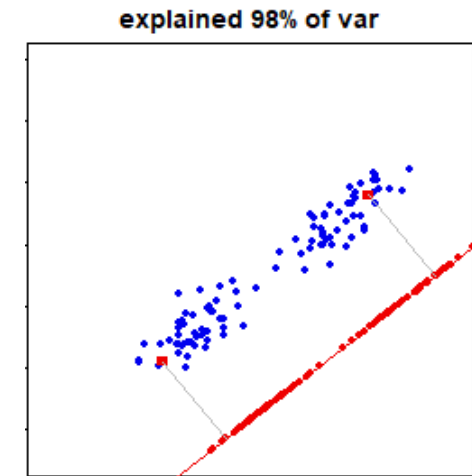
- For every covariance matrix exists a spectral decomposition

$$\Gamma \Lambda \Gamma' = \begin{pmatrix} \gamma_{11} & \gamma_{21} & \cdots & \gamma_{p1} \\ \gamma_{12} & \gamma_{22} & \cdots & \gamma_{p2} \\ \vdots & \vdots & \ddots & \vdots \\ \gamma_{1p} & \gamma_{2p} & \cdots & \gamma_{pp} \end{pmatrix} \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_p \end{pmatrix} \begin{pmatrix} \gamma_{11} & \gamma_{12} & \cdots & \gamma_{1p} \\ \gamma_{21} & \gamma_{22} & \cdots & \gamma_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \gamma_{p1} & \gamma_{p2} & \cdots & \gamma_{pp} \end{pmatrix}$$

- $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$ – the matrix with **eigenvalues**.
- The order does not matter, so let's order them from the largest to the smallest: $\lambda_1 \geq \dots \geq \lambda_p$
- $\Gamma = (\gamma_1, \dots, \gamma_p)$ – the matrix with eigenvectors
- Note:
 - $\text{rank}(\text{cov}(X)) = \#\{\lambda_i \neq 0\}$
 - $\sum_{i=1}^p \text{var}(X_i) = \sum_{i=1}^p \lambda_i$
 - Vectors γ_i are of length 1 ($\gamma_i' \gamma_i = 1$) and orthogonal ($\gamma_i' \gamma_j = 0, i \neq j$).

PCs

- Recall the optimization problem: $\sum \text{var}(\beta_j X_j) \rightarrow \max_{||\beta||=1}$
- The transformed (not yet reduced) dataset Y (called the PCs) is:
 - The first PC $Y_1 = \gamma_1'(X - \bar{X})$
 - ...
 - The p -th PC $Y_p = \gamma_p'(X - \bar{X})$
 - In matrix form $Y = \Gamma'(X - \bar{X})$
- Note:
 - Recall:** the shift unimportant.
 - PCs have zero mean: $\bar{Y} = 0$
 - Their $\text{cov}(Y) = \Lambda = \begin{pmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_p \end{pmatrix}$
 - Dimension reduction: keep only the first few PCs that explain most of the variance.



How many PCs?

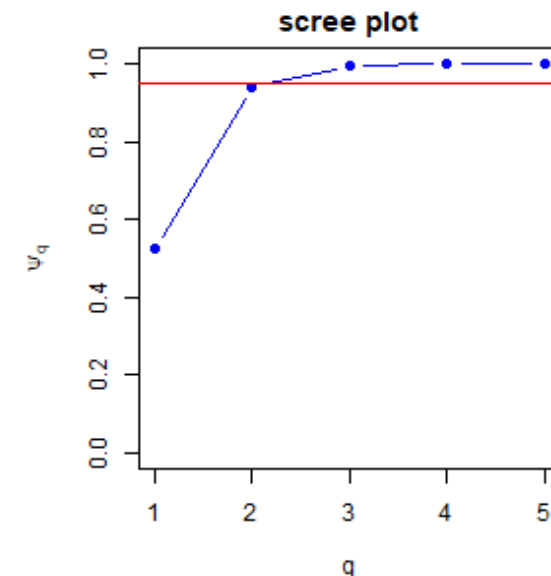
- The first PC1 is the most relevant (it has the largest variance λ_1), then the second, and so on.
- The explained variance by the first q PCs

$$\psi_q = \frac{\lambda_1 + \dots + \lambda_q}{\lambda_1 + \dots + \lambda_p}, q \in \{1, \dots, p\}$$

- **Note:** $\psi_q \in (0, 1]$, increases with q and $\psi_p = 1$.
- The scree plot for decision about the number of effective q .
- Choice of q (reduced dimension):
 - a) explained variance threshold (e.g., 90% or 95%),
 - b) elbow method.

- **Example:**

$\lambda_1 = 10$	$\psi_1 = 0.52$
$\lambda_2 = 8$	$\psi_2 = 0.94$
$\lambda_3 = 1$	$\psi_3 = 0.99$
$\lambda_4 = 0.1$	$\psi_4 = 1.00$
$\lambda_5 = 0$	$\psi_5 = 1.00$

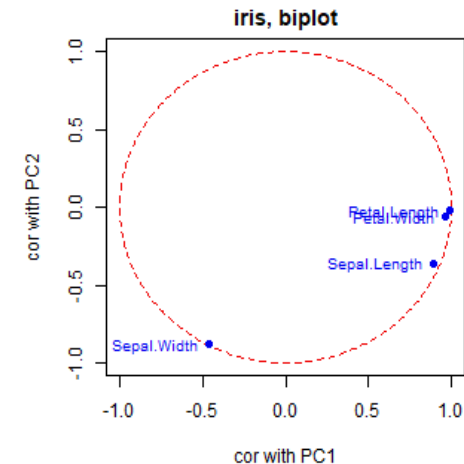
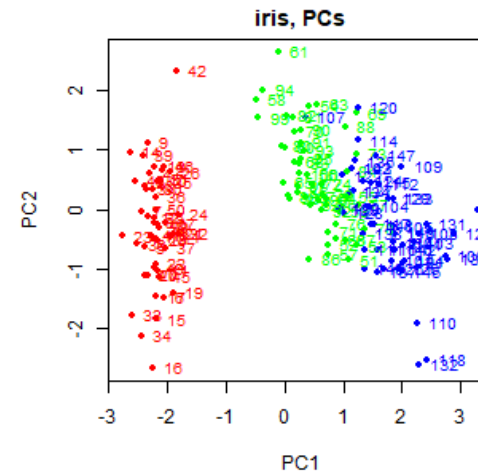
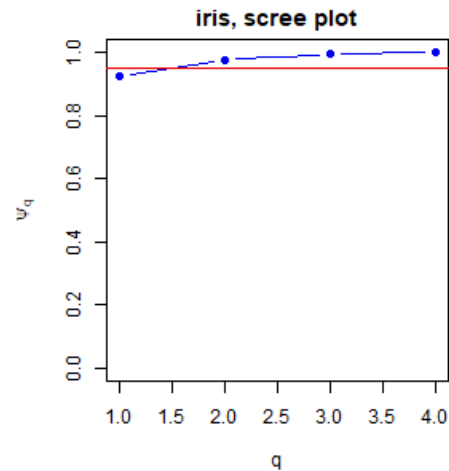
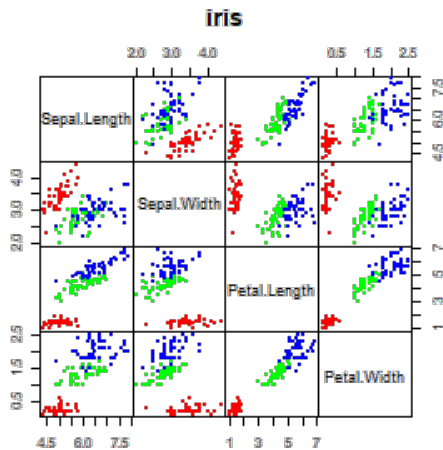


Visualization in PCA

1. Data itself
2. Scree plot
3. PCs
4. Biplot: correlation $\rho_{ij} = \text{cor}(X_i, Y_j)$ measures how good the j -th PC Y_j explains X_i

Note: $\sum_{j=1}^p \rho_{ij}^2 = 1 \Rightarrow \sum_{j=1}^q \rho_{ij}^2 \leq 1 \Rightarrow \text{biplot}$

Example (iris data):



Summary for PCA

Algorithm:

1. covariance matrix $\text{cov}(X) = (X - \mu)'(X - \mu)$;
2. eigenvalues $\lambda = (\lambda_1, \dots, \lambda_p)$ and eigenvectors $\Gamma = (\gamma_1, \dots, \gamma_p)$;
3. PCs $Y_j = \gamma_j'(X - \bar{X})$;
4. the scree plot for a decision about the number of effective q ;
5. visualization: data, scree plot, PCs, biplot;
6. take for further analysis only first q PCs.

Advantages:

- easy calculation technique, explicit formulas,
- easy to interpret,
- asymptotic convergence.

Disadvantages:

- suitable only for data with linear dependent variables (nonlinear $\Rightarrow$ e.g., kernel PCA, autoencoder)
- sensitive to outliers;
- only for continuous variables.

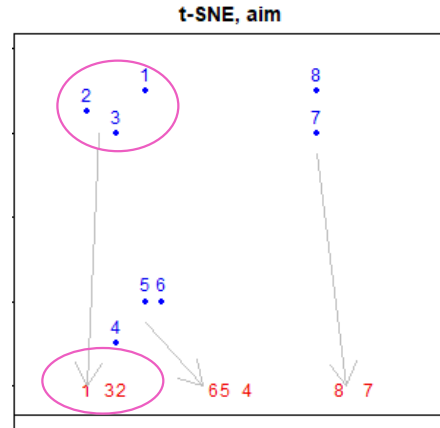
1.2. t-SNE: Idea

Aim: dimension reduction in the sense that points close together in the original space will also be close together in the lower dimension.

Example (image of size 64×64):

An image can be considered as a row with $64 \times 64 \times 3 = 12\,288$ elements.

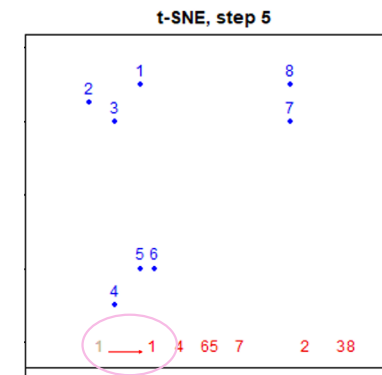
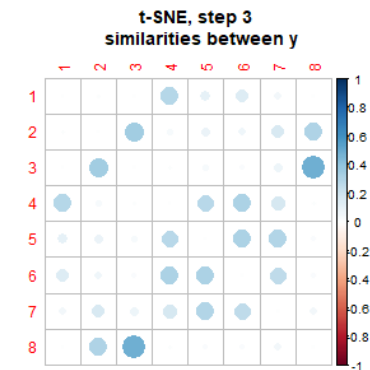
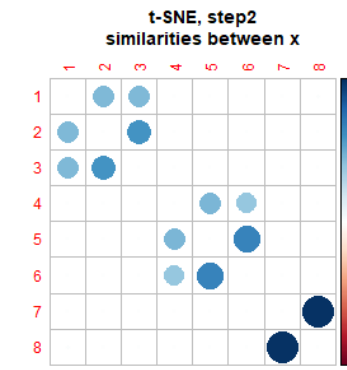
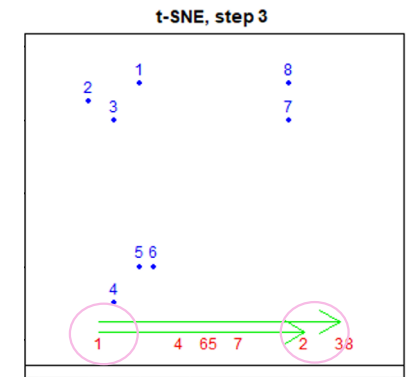
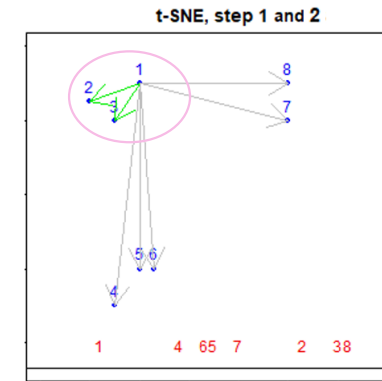
Algorithm Idea:



Algorithm

1. Drop the points to a lower dimension (often 2 or 3), randomly : $x_i \rightarrow y_i$
2. Calculate matrix P with "similarities" between the original points: $\exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma_i^2}\right)$, scale them, average;
3. Calculate the matrix Q with "similarities" in the new dimension: $(1 + \|y_i - y_j\|^2)^{-1}$, scale them, average;
4. Kullback-Leibler divergence between P and Q :

$$D_{\text{KL}}(P||Q) = \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}}$$
5. Move points to minimize the KL divergence, return to step 3.



Example for iris data and Summary

Example (iris data):

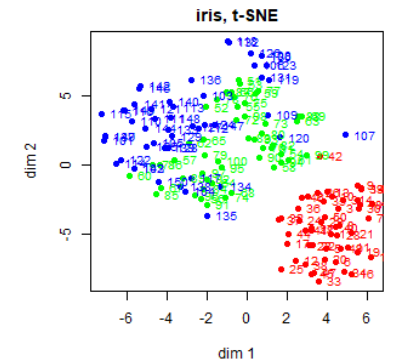
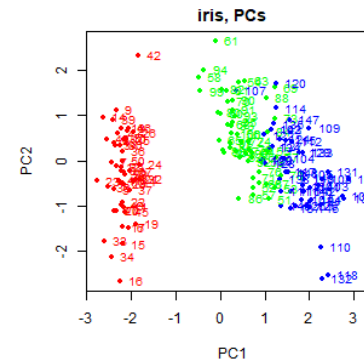
- Why is PCA better than t-SNE?
- The variables are strongly linear correlated
- $\Rightarrow$ PCA is so good

Advantages

- works for any dependency between variables,
- also used for clustering.

Disadvantages

- computational intensive,
- every time different results because of stochastic steps,
- is not applicable for a new data point,
- only for continuous variables.



1.3 Autoencoder

Idea

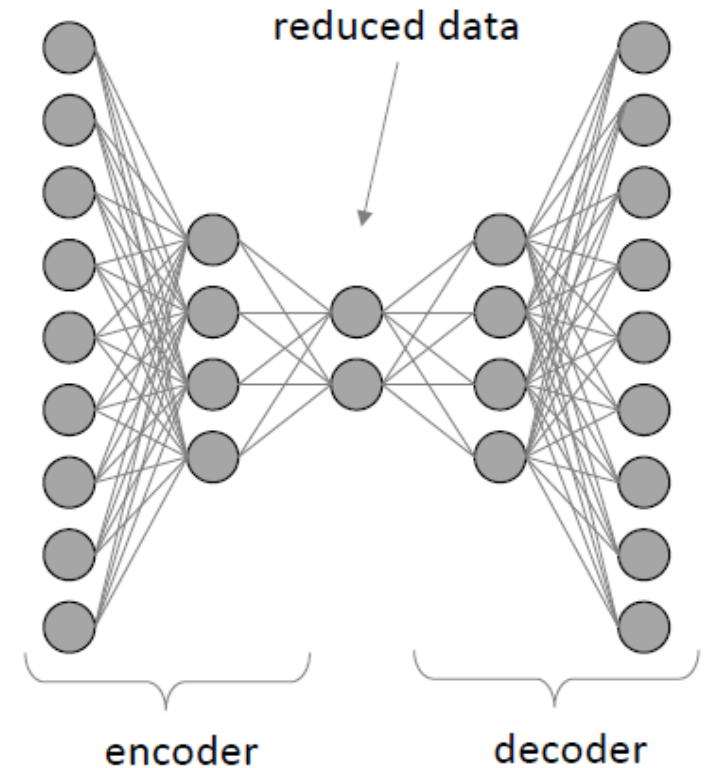
- train a model (a NN) to reduce the data dimension and then increase back to get data close to the original,
- if the model is successful then the layer in the middle contains the whole information on the original data.

Advantages

- Applicable for all data scales,
- Applicable also for out-of-sample.

Disadvantages

- Computationally intensive,
- Works only for objects similar to objects in the training data.



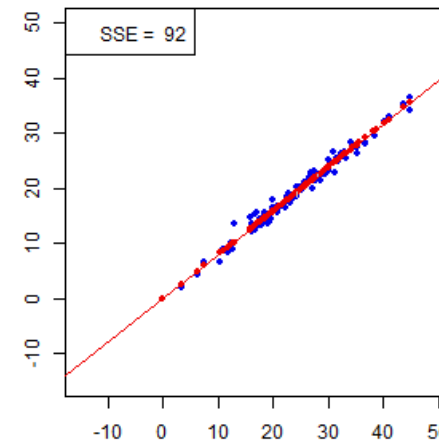
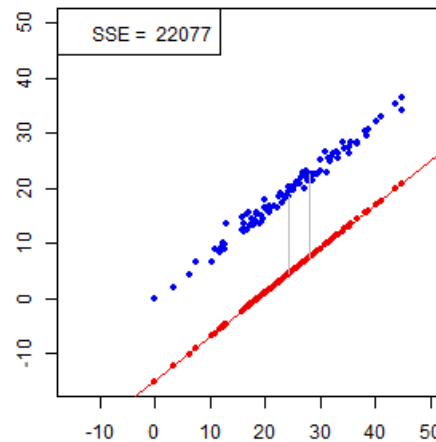
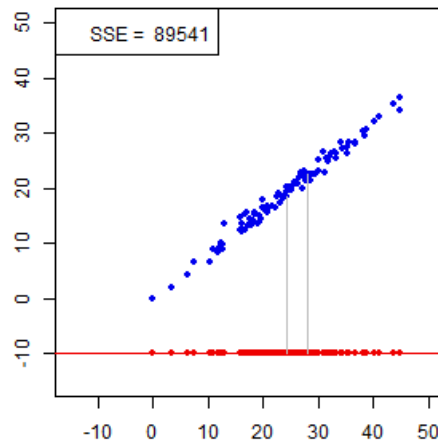
2. Regression

Aim of any regression: Model the relationship between the **target** (dependent, regressand, output) variable and **other** (independent, regressors, input, features) **variables**.

$$Y \approx m(X)$$

- The task can be prediction, explanation, or control.
- And “**linear**” stays for a linear relationship

$$y_i \approx \beta_0 + \beta_1 X_{i,1} + \dots + \beta_p X_{i,p}, \quad i = 1, \dots, n$$



Questions: Is the slope important? Is the shift important?

Multiple Linear Regression

The model

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \varepsilon_i, i = 1, \dots, n$$

with

- parameters $\beta = (\beta_0, \beta_1, \dots, \beta_p)'$, to estimate.
- errors $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n)'$, should be as small as possible
 - $E(\varepsilon) = 0$ and
 - $\text{Var}(\varepsilon) = \sigma^2 I$ (i.i.d. – independent and identically distributed),

Note:

- if $\varepsilon \sim N(0, \sigma^2 I)$ then $y \sim N(X\beta, \sigma^2 I)$ and prediction $y_0 \sim N(x_0' \beta, \sigma^2)$.

In matrix form

$$Y = X\beta + \varepsilon$$

Estimation

- Model: $Y = X\beta + \varepsilon$
- Errors: $\varepsilon = Y - X\beta$
- OLS estimator: $\|y - X\beta\|^2 \rightarrow \min_{\beta} \Rightarrow \hat{\beta} = (X'X)^{-1}X'Y$
- Prediction $\hat{Y}_0$ for given x_0 : $\hat{Y}_0 = x_0\hat{\beta} = x_0(X'X)^{-1}X'Y$
- Note:
 - $\text{Cov}(\hat{Y}_0) = \sigma^2\{I + x_0'(X'X)^{-1}x_0'\}$
 - $\text{Cov}(\hat{\beta}) = \sigma^2(X'X)^{-1}$
- Problem: calculation of $(X'X)^{-1}$ can be problematic in cases
 - X of large size, or
 - variables in X are linear dependent.

Goodness of Fit

- **Parameter significance:** (required assumption $\varepsilon \sim N(0, \sigma^2 I)$)
- if p -value of a test for $H_{0j}: \beta_j = 0$ is small (e.g. $p \leq 0.05$) $\Rightarrow$ the parameter β_j is significant.

- **Coefficient of determination:**
$$R^2 = 1 - \frac{SS_{residuals}}{SS_{total}} = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

- Note:

- $R^2 \in [0, 1]$
- An R^2 close to one means the model is good:

0.00–0.10	0.10–0.30	0.30–0.60	0.60–0.80	0.80–1.00
Very weak explanatory power	Weak	Moderate explanatory power	Good explanatory power	Very strong fit
The model explains only a small share of the variation.	Can still be meaningful in noisy social, economic, or behavioral data.	A reasonable fit in many applied settings.	The model captures a substantial part of the variation.	More common in controlled experiments, calibration, or physical processes.

- Dimension reduction:
 - R^2 is decreasing (nonincreasing) with excluding variables, also nonrelevant.
 - $\Rightarrow$ adjusted $R^2 = 1 - \frac{SS_{residuals}/(n-k)}{SS_{total}/(n-1)} \rightarrow \max$ or $AIC = \ln \hat{\sigma}^2(k) + k \frac{2}{n} \rightarrow \min$
 with $k \leq p$ – number of variables in a reduced model
 - $\Rightarrow$ backward variable selection.

Variable Transformation

- **Dummies:**

If variable x_p is categorical with d values $\Rightarrow d - 1$ dummies are required.

- In some cases, y depends linearly on some function of x .
- Variable transformations:
 - any transformation of X is possible

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 \log x_{i1} + \beta_3 x_{i1}^2 + \beta_4 x_{i1} x_{i2} + \dots + \varepsilon_i, i = 1, \dots, n$$

- but avoid transformations of Y

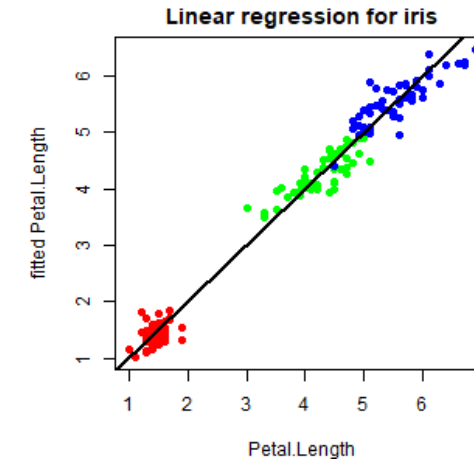
~~$$\log y_i = \beta_0 + \beta_1 x_{i1} + \dots + \varepsilon_i, i = 1, \dots, n$$~~

Interpretation

Example (iris data): $\text{Pe. Le} = \beta_0 + \beta_1 \text{Se. Le} + \beta_2 \text{Se. Wi} + \beta_3 \text{Pe. Wi} + \beta_4 (\text{Sp.} = \text{versi}) + \beta_5 (\text{Sp.} = \text{virgi}) + \varepsilon$

	β_j	p-value
(Intercept)	-1.11	0.00
Se. Le	0.61	0.00
Se. Wi	-0.18	0.03
Pe. Wi	0.60	0.00
Sp. = <i>versi</i>	1.46	0.00
Sp. = <i>virgi</i>	1.97	0.00
R^2	0.9786	
$\text{adj}R^2$	0.9778	

Model	AIC
full	-395
-Se. Wi	-391
-Pe. Wi	-373
-Sp.	-338
-Se. Le	-291



- R^2 is high $\Rightarrow$ good prediction, the same on the plot: points lie close to the black line.
- $\hat{\beta}_{\text{Se. Le}}, \hat{\beta}_{\text{Pe. Wi}} > 0 \Rightarrow \text{Pe. Le}$ is inscreasing with inscreasing Se.Le and Pe.Wi but decreasing with Se.Wi
- The virgi has the largest Pe.Le
- All p -values are small (< 0.05) $\Rightarrow$ all coefficients are significant.
- linear regression was fitted 5 times, in worst case $\frac{p(p+1)}{2}$ times,
- in this example, no reduction.

Model Reduction Techniques

- **Backward** model reduction based on AIC (but many submodels to fit);
- **PCR**: principal component regression: apply PCA to X and replace X with the first important PCs (but every variable is to observe still);
- **(adaptive) lasso**: panellation, e.g., least absolute shrinkage and selection operator (recommended)
 - OLS
$$||y - X\beta||^2 \rightarrow \min_{\beta}$$
 - Lasso:
$$||y - X\beta||^2 + \lambda ||\beta||_1 \rightarrow \min_{\beta}$$
 - λ is a hyperparameter, easy to optimize;
 - Faster and usually reduces more than backward model reduction
 - Improved Lasso: adaptive, ...

Summary for Linear Regression

- **Algorithm:**

1. preparation: visualize data, decide about variable transformation;
2. estimate the model;
3. interpret parameters, goodness of fit (R^2);
4. visualize results (prediction with KI vs Y , residuals vs. Y);
5. check the assumptions (normality, homogeneity, variable correlations);
6. decide about model reduction.

- **Advantages:**

- easy calculation technique, explicit formulas;
- easy to interpret;
- asymptotic convergence.

- **Disadvantages:**

- suitable only for data with linear dependent variables;
- sensitive to outliers;
- only for continuous variables, or dummies are required.

3. Logistic Regression

Recall – aim of any regression:

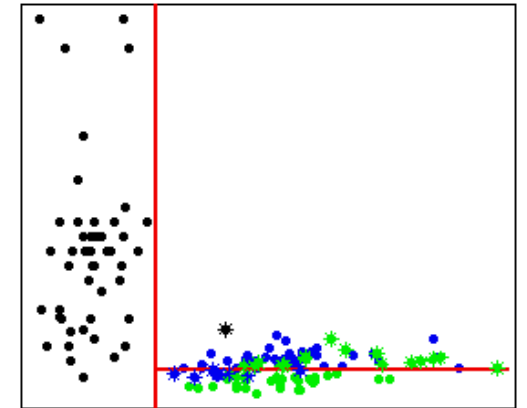
Model the relationship between the target variable and other variables.

$$Y \approx m(X)$$

- Y is categorical (classes, labels) and the output of regression is the probabilities for each class.
- **Logistic regression** solves classification problem
- **Multinomial regression** is its extension for a multiclass classification problem.

• **Example** (iris data):

- **Logistic regression** $P(\text{Sp.} = 'versi') \approx f(\text{Se. Le, Se. Wi, Pe. Le, Pe. Wi})$
- **Multinomial regression** $P(\text{Sp.}) \approx f(\text{Se. Le, Se. Wi, Pe. Le, Pe. Wi})$



Model

- Let Y have only two classes $\{0,1\}$.
- Aim: for a given x_i estimate the probabilities for $Y = 1$ and $Y = 0$.
- Model is

$$\log \frac{P(Y = 1|x_i)}{P(Y = 0|x_i)} = x_i' \beta \Leftrightarrow \frac{P(Y = 1|x_i)}{P(Y = 0|x_i)} = \exp(x_i' \beta) \Leftrightarrow P(Y = 1|x_i) = \frac{1}{1 + \exp(-x_i' \beta)} = \sigma(z)$$

with $z = x_i' \beta$.

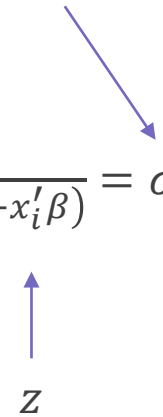
- **Note:**

- if $\beta_i > 0$ then $P(Y = 1)$ increases with x_i .

- **Prediction**

$$\hat{y}_0 = \begin{cases} 1, & \hat{P}(Y = 1) > \frac{1}{2} \\ 0, & \text{else} \end{cases}$$

sigmoid function
also called **logit**



Interpretation

Example (iris data): $P(Sp = 'versicolor') = \frac{1}{1 + \exp\{-(\beta_0 + \beta_1 \text{Se.Le} + \beta_2 \text{Se.Wi} + \beta_3 \text{Pe.Wi})\}}$

	β_j	p -value
(Intercept)	7.38	0.00
Se. Le	-0.25	0.71
Se. Wi	-2.80	0.00
Pe. Le	1.31	0.05
Pe. Wi	-2.78	0.02

- $\hat{\beta}_{Pe.Le} > 0 \Rightarrow$ the probability for “versicolor” **increases** with increasing “petal length”
- $\hat{\beta}_{Se.Le}, \hat{\beta}_{Se.Wi}, \hat{\beta}_{Pe.Wi} < 0 \Rightarrow$ the probability for “versicolor” **decreases** with increasing “sepal length”, “sepal width”, “petal width”
- $\hat{\beta}_{Se.Wi}$ is the largest in absolute value $\Rightarrow$ the probability changes faster with changing “petal width”
- p -value for sepal length is large (larger than 0.05) $\Rightarrow$ sepal length is insignificant in this model $\Rightarrow$ try dimension reduction technique to reduce the model

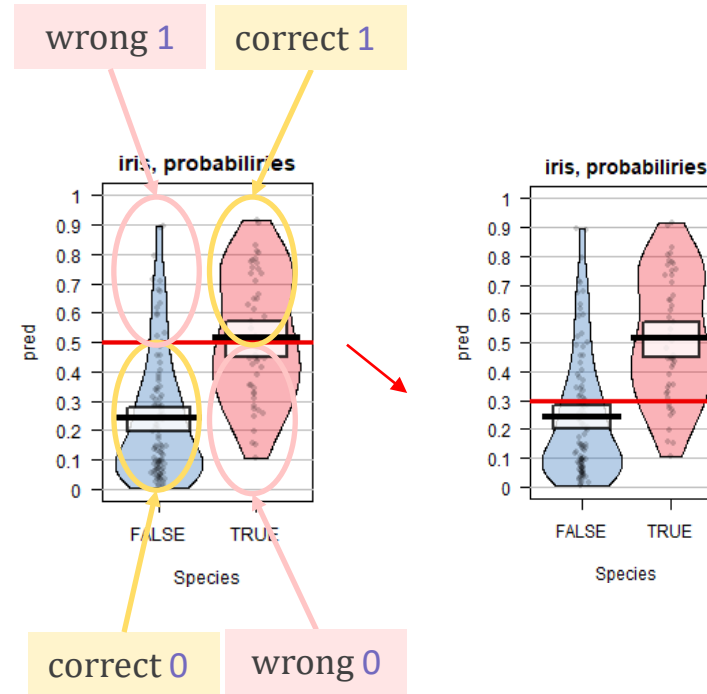
Goodness of fit

conf matrix	$\hat{Y} = 1$	$\hat{Y} = 0$	$n(Y)$
$Y = 1$	25	25	50
$Y = 0$	14	86	100
$n(\hat{Y})$	39	111	150

$$\text{accuracy} = \frac{25 + 86}{150} = 0.74$$

$$\text{sensitivity} = \frac{25}{50} = 0.50$$

$$\text{specificity} = \frac{86}{100} = 0.86$$

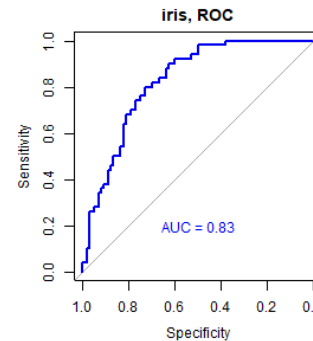


conf matrix	$\hat{Y} = 1$	$\hat{Y} = 0$	$n(Y)$
$Y = 1$	41	9	50
$Y = 0$	31	69	100
$n(\hat{Y})$	72	78	150

$$\text{accuracy} = \frac{41 + 69}{150} = 0.73$$

$$\text{sensitivity} = \frac{41}{50} = 0.82$$

$$\text{specificity} = \frac{69}{100} = 0.69$$



Summary for Logistic Regression

Advantages

- easy to implement;
- easy to interpret;
- model reduction is the same as in linear regression.

Disadvantages

- works well only if the assumption about dependency is true (similar to linear regression);
- works poorly for unbalanced data (critical in general for all classification methods)
 - e.g., what if $n(1) = 95$ and $n(0) = 5$? What if these 5 zeros are critical?